Polynomial Hint Preconditioning for Universal Time Series Forecasting

Anonymous

Abstract

Patch-based transformers for time series forecasting embed each patch independently, creating a cross-patch information bottleneck: temporal patterns spanning multiple patches must be recovered entirely through attention. We propose **polynomial hint preconditioning**, which applies a fixed Chebyshev FIR filter to the input and concatenates the result as an auxiliary channel before patching. This injects cross-patch temporal information with negligible overhead (12K parameters, 0.11%). On GIFT-Eval (97 configurations, 5 seeds), our method improves geometric mean MASE by 2.9% over the Moirai-2 Small baseline ($p < 10^{-15}$), winning 329/485 comparisons. Controlled ablations—duplicate channel *hurts* (44.7% wins), zero channel is neutral—confirm the improvement is driven by polynomial content. At 100K training steps, the benefit grows to 3.9%.

1 Introduction

Universal time series forecasting models [Woo et al., 2024, Das et al., 2024, Ansari et al., 2024] train on diverse corpora to generalize zero-shot. The Moirai family [Woo et al., 2024, Liu et al., 2024] uses patch-based transformer encoders: the input is split into non-overlapping patches of size P, each projected to a d-dimensional embedding [Nie et al., 2023, Dosovitskiy et al., 2021]. This creates a *cross-patch information bottleneck*: before the first attention layer, each embedding encodes only P time steps, so multi-patch patterns (e.g., daily cycles in hourly data spanning 1.5 patches) must be discovered through attention alone.

We propose **polynomial hint preconditioning**: before patching, we apply a fixed Chebyshev polynomial FIR filter [Oppenheim et al., 1999, Parks and McClellan, 1972] to the raw series and concatenate the result as an auxiliary input channel:

$$h[t] = T_d(B^s)\, x[t], \quad s = P, \tag{1}$$

where T_d is the degree-d Chebyshev polynomial and B^s is the backshift operator. For d=4, s=16: $h[t] = x[t] - 2x[t{-}32] + 0.5x[t{-}64]$, injecting information from 2 and 4 patches back. Only the input projection widens (12K extra parameters out of 11.4M). This is inspired by universal sequence preconditioning [Marsden and Hazan, 2024], which shows polynomial transformations reduce spectral complexity for online prediction. We provide the filtered signal as a "hint" rather than replacing the input, letting the model combine both representations.

2 Method

Hint channel. Given the normalized time series x, we compute the hint residual $\tilde{h}[t] = T_d(B^s)x[t] - x[t]$ and partition it into patches aligned with the original patching. The input to the projection becomes $3P$-dimensional: $\mathbf{e}_i = \text{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i])$, where $\mathbf{p}_i$ is the data patch, $\mathbf{m}_i$ the observation mask, and $\tilde{\mathbf{h}}_i$ the hint patch. Only the first linear layer widens; the transformer encoder is unchanged.

Hint dropout. During training, each patch's hint is independently zeroed with probability p_{drop}=0.1, preventing over-reliance on hints. At inference, the full hint is provided.

3 Experiments

Setup. We train Moirai-2 Small [Liu et al., 2024] (11.4M params, 6-layer transformer, P=16) from scratch on the LOTSA corpus [Woo et al., 2024] for 10K steps (batch 256, AdamW, cosine LR 10^{-3}, 1K warmup, bf16). Evaluation uses GIFT-Eval [Vijay et al., 2024] (97 dataset×horizon configs) with geometric mean MASE at context length 4,000. We run 5 seeds (0, 1, 2, 7, 42) and use paired sign tests pooled across seeds (485 comparisons).

Conditions. Baseline (standard Moirai-2), Zero (hint architecture with all-zero hints), Duplicate (hint = copy of input), and **HD10** (Chebyshev d=4, stride 16, 10% dropout).

3.1 Main Results

Table 1 shows that HD10 improves MASE by 2.9% over the baseline ($p < 10^{-15}$). The capacity controls confirm the polynomial content drives the improvement: a *duplicate* channel hurts (44.7% wins), while a *zero* channel is neutral (54.0%). HD10 vs. Duplicate: 336/485 wins ($p = 6 \times 10^{-18}$). HD10 wins on every seed individually (range: $+0.4\%$ to $+5.0\%$).

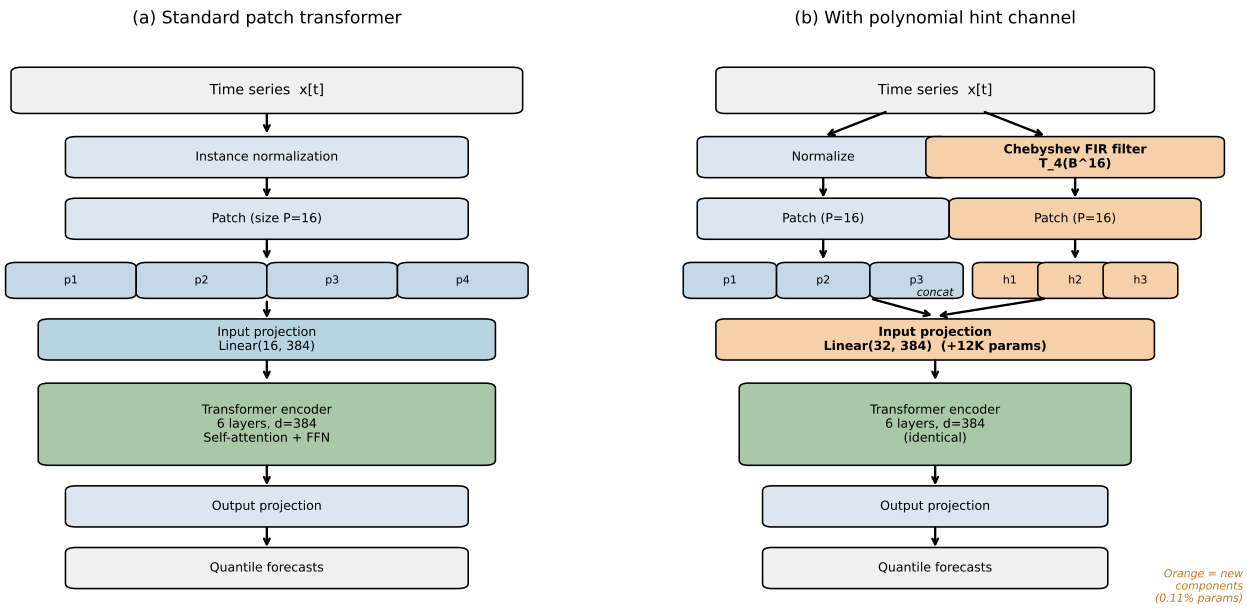

Figure 1: Overview: (a) standard patch transformer; (b) with hint preconditioning. Only the input projection is wider (orange). Total overhead: 12K parameters (0.11%).

Table 1: Main results (5 seeds, 485 comparisons on GIFT-Eval). HD10 significantly outperforms baseline and both capacity controls.

Method	Mean MASE	vs BL	Wins/485	p-value
HD10 (Ours)	**1.1765**	**−2.89%**	**329 (67.8%)**	$\mathbf{1.5 \times 10^{-15}}$
Zero channel	1.2061	−0.45%	262 (54.0%)	0.042
Baseline	1.2116	—	—	—
Duplicate channel	1.2162	+0.38%	217 (44.7%)	0.99

3.2 Ablations

Figure 2 summarizes ablations over polynomial degree (d=4 optimal), filter stride (s=P optimal—filter taps align with patch boundaries), basis type (Chebyshev > EMA > Legendre $\gg$ finite differences), and hint dropout (10% optimal, inverted-U pattern). An inference-time ablation confirms the model learns specific polynomial structure: replacing hints with zeros degrades MASE by 9.3%; random noise by 14.5%; duplicate input by 29.9%.

3.3 Extended Training and Analysis

At 100K training steps (5 seeds), HD10 improves by 3.9% over the baseline (mean MASE 1.1871 vs. 1.2351), while Duplicate (-0.9%) and Zero (-1.7%) both *hurt*. The baseline exhibits training instability at long horizons (seed variance 1.21–1.28), while HD10 stays stable (1.17–1.21). Notably, training loss differences are much smaller than eval gains: at 10K steps, HD10's loss is only 1.0% lower than baseline (0.1350 vs. 0.1364), yet eval MASE improves by 2.9%—a 2.9× amplification (2.4% vs. 3.9% at 100K). Duplicate and Zero controls have training loss comparable to baseline despite worse eval, confirming that hints act as a regularizer rather than an optimization accelerator.

Domain & frequency. The benefit is strongest in high-frequency domains: transport ($+6.9\%$, p=4×10^{-8}), energy ($+4.0\%$, p=2×10^{-4}), consistent with the filter's lookback scale (32, 64 steps = 8, 16 hours for 15-min data). Weather and healthcare show no significant improvement. The benefit increases with prediction horizon ($+1.4\%$ short $\rightarrow$ $+5.1\%$ long) and is approximately constant ($\sim$5%) across context lengths 512–4,000. HD10 at context 2,000 outperforms the baseline at context 4,000 (1.1713 vs. 1.2185).

Learning rate robustness. HD10 is insensitive to learning rate (0.1% difference between η=10^{-3} and 2×10^{-3}), while the baseline varies by 2.3%. At matched optimal LR (2×10^{-3}), HD10 still wins significantly (280/485, p=3.8×10^{-4}), though the effect is smaller ($+0.7\%$).

4 Related Work

Time series foundation models [Woo et al., 2024, Liu et al., 2024, Das et al., 2024, Ansari et al., 2024, Goswami et al., 2024] train on large corpora for zero-shot forecasting. Patch-based architectures [Nie et al., 2023, Dosovitskiy et al., 2021] are now dominant, with PatchTST using overlapping patches to partially address the cross-patch bottleneck. Our approach is orthogonal:

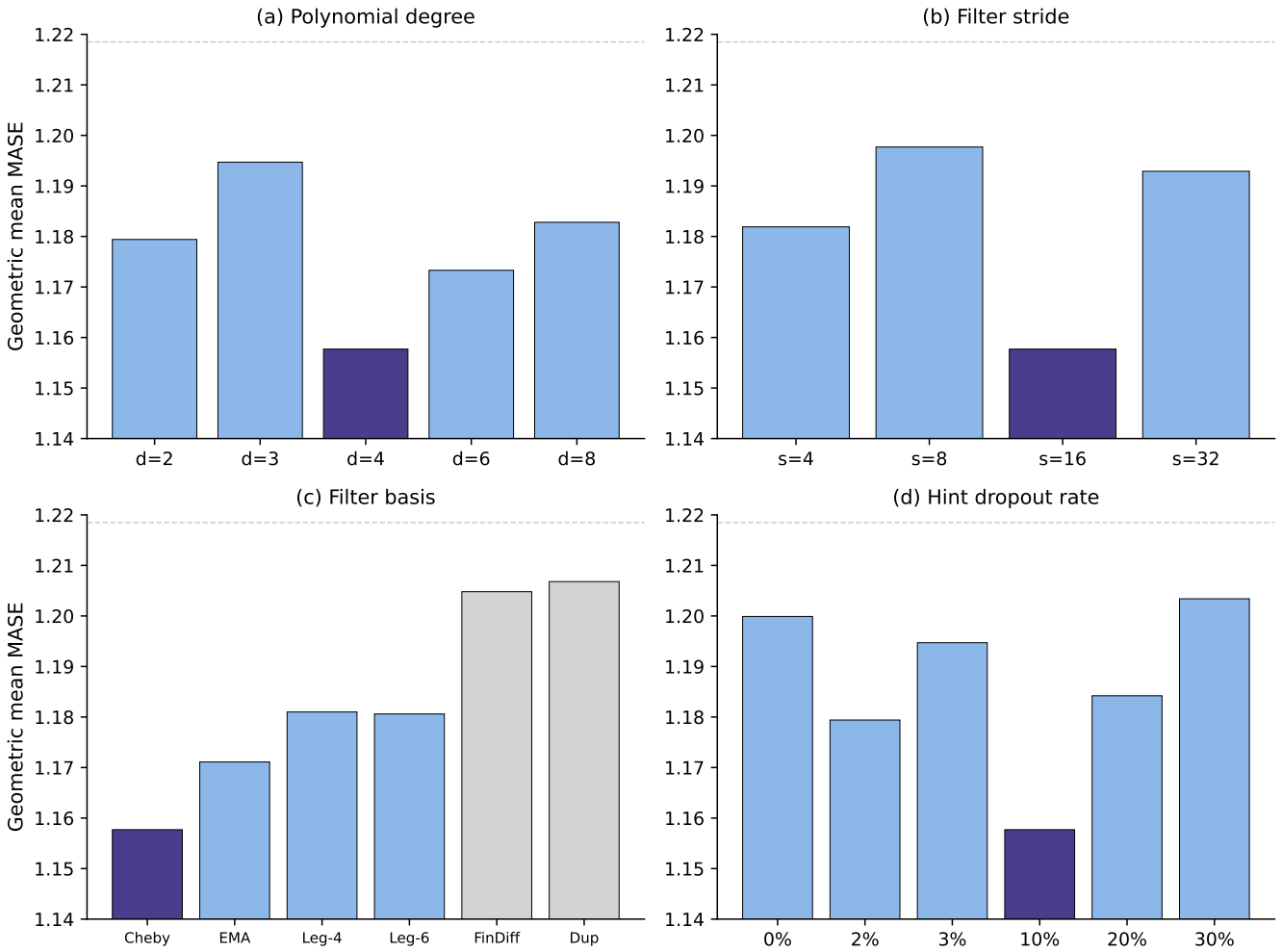

Figure 2: Ablations (seed 0, 97 configs). Bar labels: wins/97 vs. baseline. Dashed line: baseline. Orange: optimal. (a) Degree d=4 optimal. (b) Stride s=16=P optimal. (c) Chebyshev basis best. (d) 10% hint dropout optimal.

we inject cross-patch information via a fixed filter with no change to attention. The polynomial basis is motivated by universal sequence preconditioning [Marsden and Hazan, 2024], which provides theoretical guarantees for polynomial transformations in online prediction, building on spectral filtering [Hazan et al., 2017, 2018]. FIR filter design with Chebyshev polynomials is classical [Parks and McClellan, 1972, Oppenheim et al., 1999].

5 Conclusion

Polynomial hint preconditioning adds a fixed Chebyshev FIR-filtered channel to patch-based time series transformers, injecting cross-patch temporal information at 0.11% parameter overhead. On GIFT-Eval with 5 seeds, it improves MASE by 2.9% ($p < 10^{-15}$), with controlled ablations confirming the polynomial content—not extra capacity—drives the gain. The method is applicable to any patch-based architecture [Woo et al., 2024, Nie et al., 2023, Das et al., 2024] with no modification to the transformer encoder. Limitations include reduced benefit at matched optimal learning rate ($+0.7\%$), no improvement on weather/healthcare domains, and inconsistent results at larger model scales (46M parameters).

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